Disaster Recovery (RPO and RTO) - Detailed Guide

Simple step-by-step explanation with examples.

Introduction

- Disaster Recovery (DR) is the process of restoring systems, applications, and data after failures such as crashes, cyberattacks, outages, or natural disasters.
- The goal is to reduce downtime and data loss.

RPO (Recovery Point Objective)

- RPO means the maximum acceptable amount of data loss measured in time.
- If RPO is 15 minutes, the business can lose at most 15 minutes of data.
- Example: If failure happens at 10:00 AM, data should be recoverable to at least 9:45 AM.

RTO (Recovery Time Objective)

- RTO means the maximum acceptable downtime after an incident.
- If RTO is 30 minutes, service must be restored within 30 minutes.
- Example: If outage starts at 10:00 AM, system should be back by 10:30 AM.

Difference

- RPO focuses on data loss.
- RTO focuses on downtime.

Step-by-Step DR Process

- 1. Identify critical systems such as website, database, payment gateway, and email.
- 2. Define RPO and RTO for each system.
- 3. Choose backup strategy such as daily backup, snapshots, or real-time replication.
- 4. Prepare DR environment such as secondary server or cloud region.
- 5. Create a recovery runbook with exact recovery steps.
- 6. Configure monitoring and alerts.
- 7. Execute recovery during disaster.
- 8. Validate system and data.
- 9. Perform root cause analysis and improvements.

Example - E-commerce Website

- Targets: RPO = 10 minutes, RTO = 30 minutes.
- Incident at 8:00 PM: production server crashes.
- Recovery restores data to 7:55 PM and application returns by 8:25 PM.
- Targets achieved successfully.

Backup and Recovery Impact

- Daily backup gives higher possible data loss.
- Frequent snapshots reduce data loss.
- Warm standby reduces recovery time.
- Hot standby with auto failover gives fastest recovery.

Interview Answer

- RPO is the maximum acceptable data loss, and RTO is the maximum acceptable downtime. Both are key parts of disaster recovery planning.